

Problem Statement

Traditional school bell systems rely on manual operation by teachers or staff, which is time-consuming, error-prone, and inflexible. Any change in schedule requires manual intervention, making the system inefficient for modern schools.

Project Overview

The Smart School Bell System is a web-based automation solution that replaces manual bell ringing with an intelligent scheduling system. It automatically rings bells based on predefined times and allows administrators to manage schedules through a simple web dashboard.

Objectives

- Automate school bell ringing
 - Reduce manual workload for teachers
 - Provide flexible schedule management
 - Improve accuracy and punctuality in school routines
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Key Features

-  Automatic bell ringing based on time
 -  Web-based dashboard (Flask)
 -  Add /  Delete bell schedules
 -  Manual bell trigger option
 -  Live clock display
 -  Bell activity logging
 -  Persistent storage using files
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Technology Used

Python (Flask), HTML, CSS, JavaScript, Winsound, File Handling, Threading

Impact

This system reduces the need for manual bell operation, saves time for staff, and improves flexibility by allowing real-time schedule updates. It enhances school efficiency by introducing automation into daily operations.

Conclusion

The Smart School Bell System provides a simple yet powerful automation solution for schools. It improves efficiency, reduces human error, and modernizes traditional school management systems through web-based technology.

